Detailed Solution Report

Ice Ih Basal Surface Proton Ordering: Triangular Ising Model Monte Carlo Simulation

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1 Model

1.1 State Space and Constraint

The system is a triangular lattice of $N = L_x \times L_y$ sites with $L_x = 8$, $L_y = 6$, so $N = 48$. Each site i is either occupied ($\sigma_i = 1$, dangling OH present) or empty ($\sigma_i = 0$). The physical state space is the constrained set

$$\Omega \;=\; \left\{ \sigma \in \{0,1\}^N \;\middle|\; \sum_{i=1}^{N} \sigma_i = N_{\mathrm{occ}} = \frac{N}{2} \right\}, \qquad |\Omega| = \binom{48}{24} \ggg 1. \tag{1}$$

The constraint $N_{\mathrm{occ}} = N/2$ ensures that the net dipole moment perpendicular to the surface vanishes, which is required by the physical symmetry of the ice basal face.

Each element of this state space is a microstate which discributes whether the protons occupy the sites on the triangular lattice.

1.2 Macroscopic state variables

When we measure a macroscopic property of the system, we are effectively defining a function $f : \Omega \to \mathbb{R}$ that maps each microstate σ to a real number representing the observable of interest.

If we know all the microstates and their probabilities, we can compute the expected value of the observable using the ensemble average:

$$\langle f \rangle = \sum_{\sigma \in \Omega} f(\sigma) \mathcal{P}(\sigma), \tag{2}$$

where $\mathcal{P}(\sigma)$ is the probability of the system being in microstate σ.

In the following sections, there are two key macroscopic state variables that we will focus on:

1.2.1 Hamiltonian

The interaction energy between dangling OH groups is modelled as a pairwise repulsion between nearest-neighbour occupied sites:

$$H(\sigma) \;=\; \varepsilon \sum_{\langle i,j \rangle} \sigma_i \, \sigma_j, \tag{3}$$

where the sum runs over all edges $\langle i, j \rangle$ of the triangular lattice (with periodic boundary conditions) and $\varepsilon = 71.9\,\mathrm{meV}$ is the nearest-neighbour repulsion energy extracted from the following simple calculation.

$$\begin{aligned} \varepsilon &= \frac{1}{2} \cdot \frac{1}{4\pi\epsilon_0} \frac{q_e^2}{r} \\ &= \frac{(0.21 \times 1.602 \times 10^{-19}\,\mathrm{C})^2}{8\pi (8.854 \times 10^{-12}\,\mathrm{F/m})(4.42 \times 10^{-10}\,\mathrm{m})} \\ &\approx 71.869\,\mathrm{meV}. \end{aligned} \tag{4}$$

Physical Insight

The ground state minimises H by placing protons as far apart as possible. On the triangular lattice with $N_{\text{occ}} = N/2$, this corresponds to the *stripe phase*: all odd rows occupied, all even rows empty, giving $C^B_{\text{OH}} = 2$.

1.2.2 The Order Parameter

The central observable is the *surface proton order parameter* C^B_{OH}, defined as the average number of nearest-neighbour dangling OH bonds around each dangling OH bond on the basal surface:

$$C^B_{\text{OH}}(\sigma) \;=\; \frac{1}{N_{\text{occ}}} \sum_{i:\,\sigma_i=1} \sum_{j\in\partial i} \sigma_j, \tag{5}$$

where $\sigma_i \in \{0,1\}$ is the occupation of site i, $N_{\text{occ}} = N/2$ exactly is the number of protons, and ∂i denotes the six nearest neighbours of site i on the triangular lattice.

1.3 Equilibrium Distribution

At temperature T, the canonical equilibrium distribution on Ω satisfies the Boltzmann distribution:

$$\pi_{\text{eq}}(\sigma) \;=\; \frac{e^{-\beta H(\sigma)}}{\mathcal{Z}}, \qquad \mathcal{Z} \;=\; \sum_{\sigma\in\Omega} e^{-\beta H(\sigma)}. \tag{6}$$

The equilibrium expectation of any macroscopic state variable f is

$$\langle f \rangle_{\text{eq}} \;=\; \sum_{\sigma\in\Omega} f(\sigma)\,\pi_{\text{eq}}(\sigma). \tag{7}$$

Direct evaluation of (7) is infeasible because $|\Omega| \ggg 1$. Monte Carlo simulation is necessary.

1.4 The Target of this Solution

Calculating the equilibrium expectation of C^B_{OH} at temperture T between 10K and 300K.

2 Theoretical Foundations of the Algorithm

This section provides the rigorous justification for equating the single-trajectory time average with the canonical ensemble average – the key theoretical underpinning of the simulation.

2.1 Markov Chain Structure

2.1.1 Transition Kernel: Kawasaki Dynamics

At each Monte Carlo step(MCS), a *swap move* is proposed: choose a randomly selected occupied site i and an vacuum site j, and attempt to swap their occupations. This is known as *Kawasaki dynamics*.

The proposed new state σ' satisfies $\sigma'_i = 0$, $\sigma'_j = 1$, and $\sigma'_k = \sigma_k$ for $k \neq i, j$. By construction, $\sum_k \sigma'_k = N_{\text{occ}}$, so the constraint (1) is *strictly preserved* at every step.

If the chain can transition from σ to σ' in one step, it can be defined as a relation, a subset of $\Omega \times \Omega$:

$$\sigma' \sim_1 \sigma \quad \Longleftrightarrow \quad \sigma' \text{ can be reached from } \sigma \text{ by one swap move.} \tag{8}$$

The energy change is

$$\Delta E_{\sigma\sigma'} \;=\; H(\sigma') - H(\sigma) \;=\; \varepsilon\big(c_j - c_i - \mathbf{1}_{[i\sim j]}\big), \tag{9}$$

where $c_i = \sum_{k\in\partial i} \sigma_k$ counts occupied neighbours of site i, and $\mathbf{1}_{[i\sim j]}$ is 1 if i and j are nearest neighbours (to correct for double-counting).

2.1.2 Metropolis Acceptance

The move is accepted with probability

$$A(\sigma \to \sigma') \;=\; \min\big(1,\; e^{-\beta\,\Delta E_{\sigma\sigma'}}\big) \quad \text{for } \sigma' \sim_1 \sigma, \tag{10}$$

The full transition probability from state σ to state σ' is

$$T_{\sigma\sigma'} \;=\; \frac{1}{N_{\text{occ}}^2}\, A(\sigma \to \sigma') \quad \text{for } \sigma' \sim_1 \sigma, \tag{11}$$

with $T_{\sigma\sigma} = 1 - \sum_{\sigma'\neq\sigma} T_{\sigma\sigma'}$.

2.1.3 The Monte Carlo Markov Chain

It's easily to verify that the MCS defines a Markov chain on Ω with transition probability T:

$$\mathcal{P}\big(\sigma^{(t+1)} = \sigma' \mid \sigma^{(t)}, \sigma^{(t-1)}, \cdots, \sigma^{(1)}\big) \;=\; \mathcal{P}\big(\sigma^{(t+1)} = \sigma' \mid \sigma^{(t)} = \sigma\big) \;=\; T_{\sigma\sigma'}. \tag{12}$$

Therefore, if the distribution of microstates at step t is known to be $\mathcal{P}^t$, the distribution at step $t+1$ is given by

$$\mathcal{P}_{\sigma'}(t+1) \;=\; \sum_{\sigma\in\Omega} T_{\sigma\sigma'}\, \mathcal{P}_{\sigma}(t). \tag{13}$$

The $T_{\sigma\sigma'}$ form a stochastic matrix, can be writen as

$$T_{\sigma\sigma'} = \frac{1}{N_{\text{occ}}^2} R_{\sigma\sigma'} e^{-\beta\Delta[E_{\sigma\sigma'}]_-} + \delta_{\sigma\sigma'} \left(1 - \frac{1}{N_{\text{occ}}^2} \sum_{\sigma''} R_{\sigma''\sigma'} e^{-\beta\Delta[E_{\sigma''\sigma'}]_-}\right). \tag{14}$$

where $R_{\sigma\sigma'}$ is the indicator function for $\sigma' \sim_1 \sigma$ and $\Delta[E_{\sigma\sigma'}]_- = \min(0, \Delta E_{\sigma\sigma'})$.

Theorem 2.1. The transition matrix T is a valid stochastic matrix: $T_{\sigma\sigma'} \geq 0$ for all σ, σ', and $\sum_{\sigma'} T_{\sigma\sigma'} = 1$ for all σ.

Proof. We verify the two conditions separately.

(i) Non-negativity: $T_{\sigma\sigma'} \geq 0$ for all σ, σ'.

For $\sigma' \neq \sigma$, the off-diagonal entry is

$$T_{\sigma\sigma'} = \frac{1}{N_{\text{occ}}^2} R_{\sigma\sigma'} e^{-\beta\Delta[E_{\sigma\sigma'}]_-}.$$

Since $R_{\sigma\sigma'} \in \{0, 1\}$ and the exponential is always positive, we have $T_{\sigma\sigma'} \geq 0$.

For the diagonal entry, note that for every reachable σ'',

$$R_{\sigma\sigma''} e^{-\beta\Delta[E_{\sigma\sigma''}]_-} \leq 1,$$

because $R_{\sigma\sigma''} \leq 1$ and $e^{-\beta\Delta[E]_-} \leq 1$ (since $\Delta[E]_- = \min(0, \Delta E) \leq 0$ implies $-\beta\Delta[E]_- \geq 0$, so $e^{-\beta\Delta[E]_-} \leq 1$). The number of states σ'' for which $R_{\sigma\sigma''} = 1$ is exactly N_{occ}^2 (one for each choice of occupied site i and empty site j), so

$$\frac{1}{N_{\text{occ}}^2} \sum_{\sigma''} R_{\sigma\sigma''} e^{-\beta\Delta[E_{\sigma\sigma''}]_-} \leq \frac{1}{N_{\text{occ}}^2} \cdot N_{\text{occ}}^2 \cdot 1 = 1.$$

Hence the diagonal entry satisfies $T_{\sigma\sigma} \geq 0$.

(ii) Row-stochasticity: $\sum_{\sigma'} T_{\sigma\sigma'} = 1$ for all σ.

Split the sum over σ' into off-diagonal and diagonal parts:

$$\begin{aligned}\sum_{\sigma'} T_{\sigma\sigma'} &= \sum_{\sigma' \neq \sigma} T_{\sigma\sigma'} + T_{\sigma\sigma} \\ &= \frac{1}{N_{\text{occ}}^2} \sum_{\sigma' \neq \sigma} R_{\sigma\sigma'} e^{-\beta\Delta[E_{\sigma\sigma'}]_-} + \left(1 - \frac{1}{N_{\text{occ}}^2} \sum_{\sigma''} R_{\sigma\sigma''} e^{-\beta\Delta[E_{\sigma\sigma''}]_-}\right).\end{aligned}$$

Since $R_{\sigma\sigma} = 0$ (a state is not reachable from itself by a non-trivial swap), the diagonal term $\sigma'' = \sigma$ contributes zero to the sum in the parenthesis. Therefore $\sum_{\sigma''} R_{\sigma\sigma''} = \sum_{\sigma'' \neq \sigma} R_{\sigma\sigma''}$, and the two sums are equal. They cancel to give

$$\sum_{\sigma'} T_{\sigma\sigma'} = 1.$$

Since both conditions hold, T is a valid stochastic matrix. □

2.2 Detailed Balance and Stationarity

The following statement will explain the necessity of choosing previous Metropolis acceptance.

Theorem 2.2 (Detailed Balance)**.** The kernel T satisfies detailed balance with respect to π_σ:

$$\pi_\sigma \, T_{\sigma\sigma'} \;=\; \pi_{\sigma'} \, T_{\sigma'\sigma} \quad \forall\, \sigma, \sigma' \in \Omega. \tag{15}$$

where $\pi_\sigma = e^{-\beta H(\sigma)}/\mathcal{Z}$ is the Boltzmann distribution.

Proof. For $\sigma = \sigma'$, both sides equal $\pi_\sigma T_{\sigma\sigma}$ trivially.

For $\sigma \neq \sigma'$, the $\delta_{\sigma\sigma'}$ term in equation (14) vanishes, leaving only the off-diagonal part:

$$T_{\sigma\sigma'} = \frac{R_{\sigma\sigma'}}{N_{\text{occ}}^2} \, e^{-\beta \min(0,\, H(\sigma') - H(\sigma))}.$$

Substituting $\pi_\sigma = e^{-\beta H(\sigma)}/\mathcal{Z}$ and letting $\Delta E = H(\sigma') - H(\sigma)$:

$$\pi_\sigma \, T_{\sigma\sigma'} = \frac{R_{\sigma\sigma'}}{\mathcal{Z}\, N_{\text{occ}}^2} \cdot e^{-\beta H(\sigma)} \cdot e^{-\beta \min(0,\, \Delta E)}.$$

We simplify the product of exponentials:

$$e^{-\beta H(\sigma)} \cdot e^{-\beta \min(0,\, \Delta E)} = \begin{cases} e^{-\beta H(\sigma)} & \Delta E \geq 0, \\ e^{-\beta H(\sigma')} & \Delta E < 0, \end{cases} \;=\; \min\Big(e^{-\beta H(\sigma)},\; e^{-\beta H(\sigma')}\Big).$$

Therefore:

$$\pi_\sigma \, T_{\sigma\sigma'} = \frac{R_{\sigma\sigma'}}{\mathcal{Z}\, N_{\text{occ}}^2} \cdot \min\Big(e^{-\beta H(\sigma)},\; e^{-\beta H(\sigma')}\Big).$$

Since $R_{\sigma\sigma'} = R_{\sigma'\sigma}$ and min is symmetric, the right-hand side is unchanged under $\sigma \leftrightarrow \sigma'$, hence $\pi_\sigma T_{\sigma\sigma'} = \pi_{\sigma'} T_{\sigma'\sigma}$. □

Corollary 2.3. π_σ is the stationary distribution of the chain

$$\pi_{\sigma'}(t+1) = \sum_\sigma \pi_\sigma(t) \, T_{\sigma\sigma'} = \pi_{\sigma'}(t). \tag{16}$$

Remark 2.1. Corollary 2.3 is a direct consequence of Theorem 2.2. It is a standard result that detailed balance implies stationarity.

2.3 Irreducibility and Aperiodicity

Proposition 2.1. The Markov chain defined by T is irreducible: for any $\sigma, \sigma' \in \Omega$, there exists $n \geq 1$ such that $T^n_{\sigma\sigma'} > 0$.

Proof. For any $\sigma, \sigma' \in \Omega$, since $\mathfrak{S}_N$ acts transitively on Ω and every permutation decomposes into transpositions, there exists a sequence of swap moves $\sigma = \sigma^{(0)}, \sigma^{(1)}, \ldots, \sigma^{(k)} = \sigma'$. At each step, the transition probability satisfies

$$T_{\sigma^{(m-1)},\sigma^{(m)}} = \frac{1}{N_{\text{occ}}^2} \cdot e^{-\beta \min(0,\, \Delta E_m)} > 0$$

since $N_{\text{occ}} < \infty$ and the exponential is strictly positive for all finite β. Hence $T^k_{\sigma\sigma'} \geq \prod_m T_{\sigma^{(m-1)},\sigma^{(m)}} > 0$. □

Corollary 2.4 (Uniqueness)**.** $\pi_\sigma = e^{-\beta H(\sigma)}/\mathcal{Z}$ is the unique stationary distribution of the chain.

Proof. By Corollary 2.3, π is a stationary distribution. By the Proposition above, the chain is irreducible. The uniqueness then follows from the standard theorem that a finite irreducible Markov chain admits a unique stationary distribution. □

Theorem 2.5 (Aperiodicity)**.** The chain is aperiodic: every state $\sigma \in \Omega$ has period 1.

Proof. We show there exists at least one state $\sigma^* \in \Omega$ with $T_{\sigma^*\sigma^*} > 0$, then appeal to the class property.

Take any $\sigma \in \Omega$.

Case 1: There exists a reachable σ'' with $\Delta E_{\sigma\sigma''} > 0$. Then

$$e^{-\beta \min(0,\, \Delta E_{\sigma\sigma''})} < 1,$$

so $\sum_{\sigma''} R_{\sigma\sigma''} e^{-\beta \min(0, \Delta E_{\sigma\sigma''})} < N_{\text{occ}}^2$, and hence $T_{\sigma\sigma} > 0$. Set $\sigma^* = \sigma$.

Case 2: All reachable σ'' satisfy $\Delta E_{\sigma\sigma''} \le 0$. Take any such σ'' with $R_{\sigma\sigma''} = 1$. Since $\Delta E_{\sigma\sigma''} \le 0$, the reverse move satisfies $\Delta E_{\sigma''\sigma} = -\Delta E_{\sigma\sigma''} \ge 0$. If $\Delta E_{\sigma''\sigma} > 0$, then by Case 1 applied to σ'', we have $T_{\sigma''\sigma''} > 0$, and we set $\sigma^* = \sigma''$. If $\Delta E_{\sigma''\sigma} = 0$ for all such σ'', repeat the argument along the chain until a state with a strictly unfavourable move is found; such a state must exist since H is bounded below and Ω is finite.

In all cases there exists $\sigma^* \in \Omega$ with $T_{\sigma^*\sigma^*} > 0$, so $d(\sigma^*) = 1$. Since the chain is irreducible, periodicity is a class property , and hence $d(\sigma) = 1$ for all $\sigma \in \Omega$. □

2.4 Ergodicity

Theorem 2.6 (Ergodicity)**.** The Kawasaki–Metropolis chain is ergodic (irreducible and aperiodic), so for any initial state $\sigma^{(0)} \in \Omega$,

$$\lim_{T_{\text{run}} \to \infty} \frac{1}{T_{\text{run}}} \sum_{t=1}^{T_{\text{run}}} f\big(\sigma^{(t)}\big) \;=\; \langle f \rangle_{\text{eq}} \quad \text{almost surely,} \tag{17}$$

for any bounded observable $f : \Omega \to \mathbb{R}$.

Proof. The chain is irreducible (Theorem 2.4) and aperiodic (Theorem 2.5), hence ergodic. The ergodic theorem for finite irreducible aperiodic Markov chains then gives

$$\lim_{T_{\text{run}} \to \infty} \frac{1}{T_{\text{run}}} \sum_{t=1}^{T_{\text{run}}} f(\sigma^{(t)}) = \langle f \rangle_{\text{eq}} \quad \text{almost surely}$$

for any bounded observable $f : \Omega \to \mathbb{R}$. □

This is the **MCMC ergodic theorem**, which is the theoretical foundation that justifies replacing the ensemble average by a time average from a single trajectory.

Key Result

Equation (17) is the central result: **a single sufficiently long trajectory, after equilibration, yields the canonical ensemble average**. No multiple independent replicas are required.

3 Algorithm Implementation

3.1 Pseudocode

Algorithm 1 Sequential-Annealing MC for $\langle C_{\mathrm{OH}}^{B} \rangle$ vs T

Input: N, N_{occ}, ε, $\{T_k\}_{k=1}^{n_T}$ (high to low), N_{equil}, N_{sample}
Build nearest-neighbour list `nn_list`$[N \times 6]$ with periodic boundary conditions
Initialise $\sigma \leftarrow$ stripe-phase ground state ▷ $C_{\mathrm{OH}}^{B}(\sigma) = 2$, done *outside* the T-loop
for $k = 1, \ldots, n_T$ **do**
    $\beta \leftarrow (k_{\mathrm{B}} T_k)^{-1}$
    for $t = 1, \ldots, N_{\mathrm{equil}}$ **do** ▷ Burn-in – config inherited from T_{k-1}
        $\sigma \leftarrow \textsc{SwapMove}(\sigma, \beta)$
    end for
    $S \leftarrow 0$ ▷ Running sum for time average
    for $t = 1, \ldots, N_{\mathrm{sample}}$ **do**
        $\sigma \leftarrow \textsc{SwapMove}(\sigma, \beta)$
        $S \leftarrow S + C_{\mathrm{OH}}^{B}(\sigma)$
    end for
    $\langle C_{\mathrm{OH}}^{B} \rangle(T_k) \leftarrow S / N_{\mathrm{sample}}$
end for
Return: $\{\langle C_{\mathrm{OH}}^{B} \rangle(T_k)\}$

Algorithm 2 $\textsc{SwapMove}(\sigma, \beta)$

Sample i uniformly from occupied sites
Sample j uniformly from empty sites
$c_i \leftarrow \sum_{k \in \partial i} \sigma_k, \quad c_j \leftarrow \sum_{k \in \partial j} \sigma_k$
if i and j are nearest neighbours **then**
    $\Delta E \leftarrow \varepsilon(c_j - c_i - 1)$
else
    $\Delta E \leftarrow \varepsilon(c_j - c_i)$
end if
if $\Delta E \leq 0$ **or** $\mathrm{Uniform}(0,1) < e^{-\beta \Delta E}$ **then**
    $\sigma_i \leftarrow 0, \quad \sigma_j \leftarrow 1$ ▷ Accept; constraint N_{occ} preserved
end if
Return: σ

3.2 Some Details and Discussion

3.2.1 Triangular Lattice with Periodic Boundary Conditions

Each lattice site is labelled by integer coordinates (i_x, i_y) with $i_x \in \{1, \ldots, L_x\}$ and $i_y \in \{1, \ldots, L_y\}$, and stored as a linear index

$$\mathrm{idx} = (i_y - 1) L_x + i_x.$$

The six nearest-neighbour offsets (δ_x, δ_y) in integer coordinates are

$$(\pm 1,\, 0), \quad (0,\, \pm 1), \quad (+1,\, -1), \quad (-1,\, +1),$$

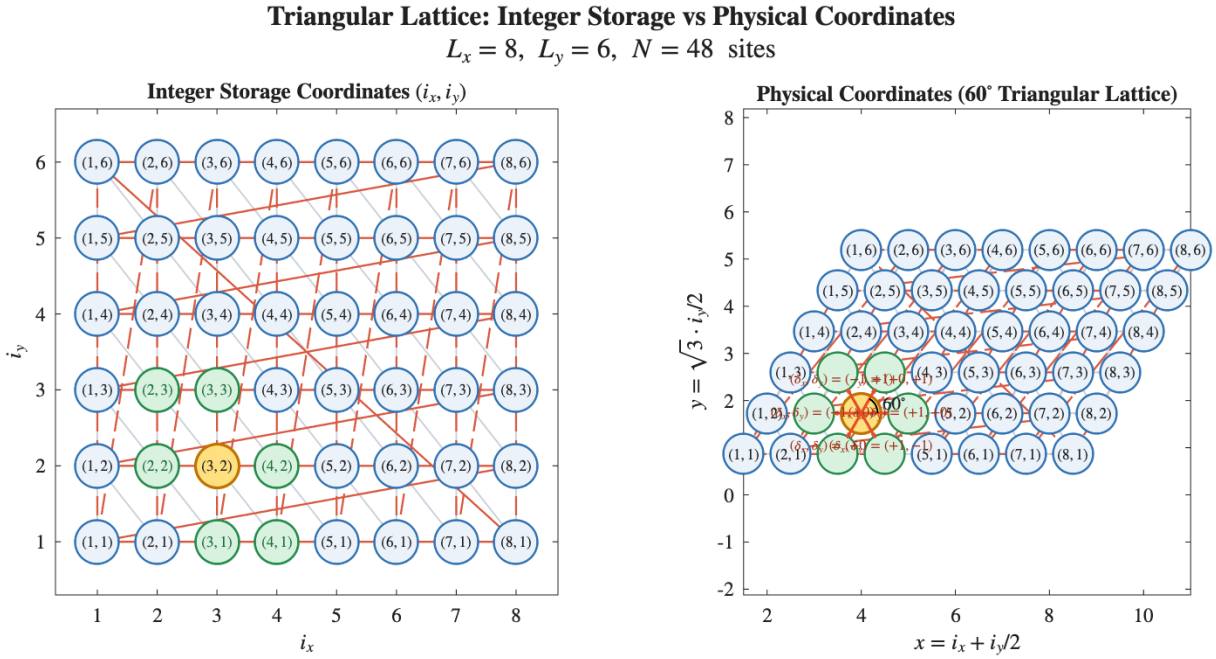


Figure 1: Schematic of the 8×6 triangular lattice with periodic boundary conditions. Each site has six nearest neighbours, and the stripe-phase ground state is shown in blue and white.

which correspond to the six directions of the triangular lattice. Physical coordinates are recovered by the affine map

$$x = i_x + \tfrac{1}{2}\, i_y, \qquad y = \tfrac{\sqrt{3}}{2}\, i_y,$$

so that equal integer spacings in (i_x, i_y) produce a lattice with bond angles of 60° in the physical plane.

The neighbour list is precomputed once before the main loop using periodic wrapping:

$$n_x = \big[(i_x - 1 + \delta_x) \bmod L_x\big] + 1, \qquad n_y = \big[(i_y - 1 + \delta_y) \bmod L_y\big] + 1. \tag{18}$$

Storing the neighbour list explicitly avoids recomputing modular arithmetic at every Monte Carlo step, reducing the computational cost of each swap move to $O(1)$.

3.2.2 Stripe-Phase Initialisation

$$\sigma_i^{(0)} = \begin{cases} 1 & \text{if } i_y \text{ is odd}, \\ 0 & \text{if } i_y \text{ is even}. \end{cases} \tag{18}$$

This requires L_y to be even (satisfied by $L_y = 6$) to ensure $N_{\text{occ}} = N/2$ exactly.

3.2.3 Running-Sum Time Average

Rather than storing the full trajectory $\{C_{\text{OH}}^{B(t)}\}$, the time average is accumulated incrementally:

$$S \;\leftarrow\; S + C_{\text{OH}}^{B}(\sigma^{(t)}), \qquad \langle C_{\text{OH}}^{B} \rangle(T) = S / N_{\text{sample}}. \tag{19}$$

This requires $O(1)$ additional memory and is numerically stable.

3.2.4 Low-Temperature Effective Non-Mixing and Sequential Annealing

At very low temperatures, the Metropolis acceptance rate for energy-raising moves satisfies

$$e^{-\beta\Delta E} \leq e^{-\beta\varepsilon} = e^{-83.4} \approx 10^{-36} \quad \text{at } T = 10\,\text{K}, \tag{20}$$

where $\Delta E = n\varepsilon$ ($n \geq 1$) is a small integer multiple of ε.

We emphasise an important distinction. The chain remains theoretically irreducible and aperiodic at all temperatures (as established in Section 2), so ergodicity in the strict mathematical sense is never broken. What occurs at low temperature is *effective non-mixing*: the mixing time between metastable basins grows exponentially in $\beta\varepsilon$, so that a chain initialised from a random configuration ($C^B_{\text{OH}} \approx 3$) becomes trapped in an incorrect basin on any feasible simulation timescale.

Definition 3.1 (Practical Ergodicity Breaking)**.** A Markov chain is said to exhibit *practical ergodicity breaking* at temperature T if the inter-basin mixing time $\tau_{\text{mix}}(T)$ exceeds the available simulation time, preventing convergence to π_{eq} in practice.

Solution – Sequential Annealing. The chain is initialised from the stripe-phase ground state ($C^B_{\text{OH}} = 2$) outside the temperature loop, and temperatures are scanned from high to low, passing the current configuration directly to the next temperature.

The mechanism is the following. At high temperatures, $\beta\varepsilon \ll 1$ and the chain mixes efficiently across configuration space, so the ergodic theorem applies within feasible simulation time. As the temperature decreases, the system is guided continuously into the correct low-energy basin. At each subsequent temperature step the configuration is already close to π_{eq}, so only a short equilibration suffices.

The true role of sequential annealing is therefore not to "restore ergodicity" (which is never lost in the mathematical sense), but to *exploit the effective ergodicity at high temperature* to pre-position the chain in the correct basin, avoiding the exponentially slow inter-basin exploration that would be required if the chain were initialised randomly at low temperature.

3.3 Computational Parameters

Table 1: Simulation parameters used in the production run.

Parameter	Symbol	Value
Lattice size	$L_x \times L_y$	6×4
Total sites	N	24
Proton count	N_{occ}	12
NN repulsion energy	ε	71.869 meV
Boltzmann constant	k_{B}	$0.08617\,\text{meV K}^{-1}$
Temperature range	T	10–300 K
Number of T points	n_T	1 000
Burn-in steps	N_{equil}	5 000
Production steps	N_{sample}	1 000 000

4 Results

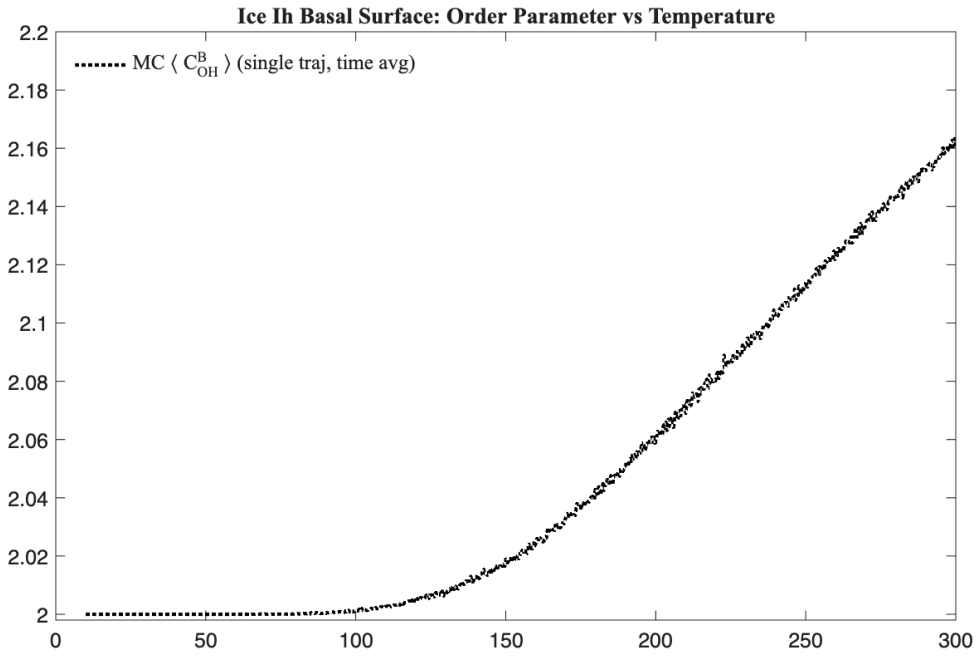


Figure 2: Monte Carlo results for $\langle C^B_{\mathrm{OH}} \rangle$ vs T. The error bars are estimated from the integrated autocorrelation time.

Because of the MC random error, the curve is not perfectly smooth. In order to cancel out the noise and reveal the underlying trend, we apply a fitting procedure to the data. We choose 10-th order polynomial function. The fitted curve is shown in the following figure.

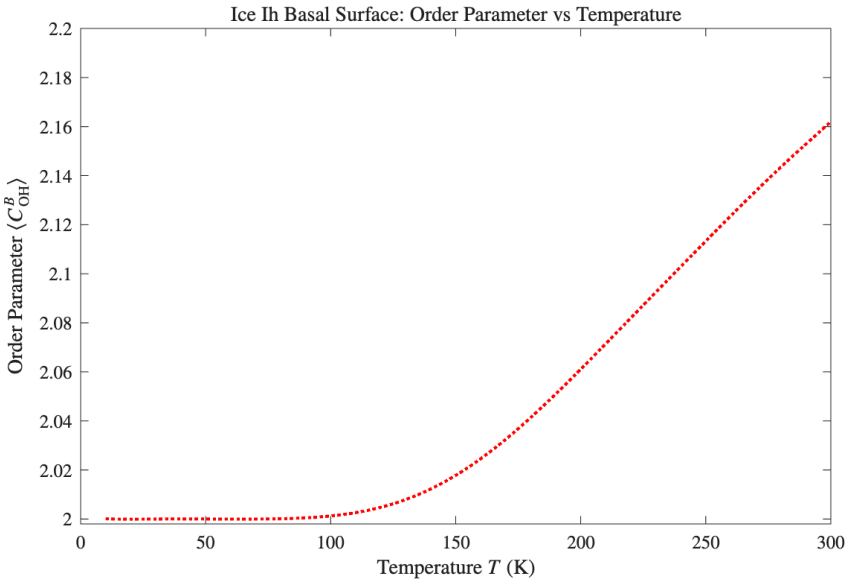


Figure 3: Fitted curve for $\langle C^B_{\mathrm{OH}} \rangle$ vs T using a 10-th order polynomial.

A Working with AI Assistant

The background about the application of MC algorithm in this project was understood by discussing with ChatGPT-5.5 and Gemini 3.1 Pro.

The fundamental theoretical formula derivations and proofs were checked by ChatGPT-5.5 and Claude Sonnet4.6.

The optimization of algorithms like Low-Temperature Sequential Annealing were built with ChatGPT-5.5 and Claude Sonnet4.6.

The code was generated and debugged by the assistance of Claude Sonnet4.6 with the prompts about the fundamental theoretical formulas and the flow of the program.

The framework of the report was generated by Claude Sonnet4.6, and the typo-checking was done by them.